US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12405534
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Payne; Alexander et al.

MEMs based spatial light modulators with improved thermal control

Abstract

Microelectromechanical systems (MEMS) based spatial light modulators (SLMs) enclosed in a package filled with a gas to enhance the reliability and lifetime of the SLM, and methods for operating the same in various applications are described. Generally, the SLM includes a number of MEMS modulators, each including a number of light reflective surfaces, at least one light reflective surface coupled to an electrostatically deflectable element suspended above a substrate, and each adapted to reflect and modulate a light beam incident thereon. The package enclosing the SLM includes an optically transparent cover through which the reflective surfaces are exposed to the light beam, and a cavity is filled a low molar mass fill gas having an atomic number of two or less and a thermal conductivity of greater than 100 mW/(m.Math.K). The SLM can include electrostatically deflectable ribbons suspended over a substrate, or a linear array of two-dimensional MEMS modulators.

Inventors: **Payne; Alexander (Ben Lommond, CA), Jacob; Gregory (Santa Clara, CA)**

Applicant: **SILICON LIGHT MACHINES CORPORATION** (San Jose, CA)

Family ID: **82704543**

Assignee: **Silicon Light Machines Corporation (San Jose, CA)**

Appl. No.: **17/591834**

Filed: **February 03, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220252866 A1	Aug. 11, 2022

Related U.S. Application Data

us-provisional-application US 63146488 20210205

Publication Classification

Int. Cl.: **G03F7/00** (20060101); **B81B7/00** (20060101); **G02B26/00** (20060101); **G02B26/02** (20060101); **G02B26/08** (20060101)

U.S. Cl.:

CPC **G03F7/70291** (20130101); **B81B7/0041** (20130101); **G02B26/001** (20130101); **G02B26/023** (20130101); **G02B26/0808** (20130101); **G02B26/0841** (20130101); **G03F7/70191** (20130101); B81B2201/04 (20130101)

Field of Classification Search

CPC: G03F (7/70291); G03F (7/70191); B81B (7/0041); B81B (2201/04); G02B (26/001); G02B (26/023); G02B (26/0808); G02B (26/0841); Y02P (10/25)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5459610	12/1994	Bloom et al.	N/A	N/A
7064883	12/2005	Payne et al.	N/A	N/A
7227687	12/2006	Trisnadi et al.	N/A	N/A
7277110	12/2006	Zarem et al.	N/A	N/A
8767170	12/2013	Sano et al.	N/A	N/A
11016287	12/2020	Payne et al.	N/A	N/A
11036030	12/2020	Lee	N/A	N/A
2005/0012197	12/2004	Smith	257/687	B81B 7/0061
2008/0239454	12/2007	Ichikawa	359/290	G02B 26/0841
2009/0128935	12/2008	Nishino et al.	N/A	N/A
2018/0307039	12/2017	Payne	N/A	G02B 26/0841
2020/0049820	12/2019	Choi	N/A	G01S 7/51
2020/0247052	12/2019	Payne	N/A	B23K 26/066

OTHER PUBLICATIONS

International Search Report for International Application PCT/US22/15334 May 13, 2022. cited by applicant

Written Opinion of International Searching Authority Application PCT/US22/15334 May 13, 2022. cited by applicant

Supplementary European Search Report for Application 22750476.8-1001 / 4288824 PCT/US22/15334 Dec. 6, 2024. cited by applicant

---

Primary Examiner: Jones; James C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority under 35 U.S.C. 119(e) to U.S. Provisional Patent Application Ser. No. 63/146,488, filed Feb. 5, 2021, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The present invention relates generally to spatial light modulators (SLMs), and more particularly to a microelectromechanical systems (MEMS) based SLM enclosed in a package filled with a fill gas having a low molar mass and high thermal conductivity to enhance the reliability and lifetime of the SLM.

BACKGROUND

(2) Spatial light modulators or SLMs include an array of one or more diffractors or modulators that can control or modulate an incident beam of light in a spatial pattern that corresponds to an electrical input to the devices. The incident light beam, typically generated by a laser, can be modulated in intensity, phase, polarization or direction. Spatial light modulators are increasingly being developed for use in various applications, including display systems, optical information processing and data storage, printing, maskless lithography, 3D printing, additive manufacturing, surface modification and optical phase modulators.

(3) One type of spatial light modulator (SLM) potentially useful in the aforementioned applications is a microelectromechanical systems (MEMS) based SLM including an array of dynamically adjustable reflective surfaces or mirrors mounted over a substrate. In operation electromagnetic radiation or light from a coherent light source, such as a laser, is projected onto the array, and alignment of the mirrors is altered by electronic signals generating electrostatic forces to displace at least some of the mirrors to modulate the phase, intensity and or angle of light reflected from the array.

(4) Unfortunately, existing MEMS based SLMs cannot handle the high power lasers employed in laser processing systems for applications including cutting, marking, engraving, and 3D printing. Typically, the failure mode of these devices when exposed to high power or temperature lasers is the “Soret effect” in which atoms of a reflective metal, such as aluminum, covering reflective surfaces in the MEMS based SLM physically migrate along from a hotter to a cooler region of the ribbon. This migration of metal atoms can reduce the reflection and hence the efficiency of the SLM, and ultimately shortens useful device life.

(5) Accordingly, there is a need for a MEMS based SLM with improved thermal and high-power handling capabilities.

SUMMARY

(6) A microelectromechanical systems (MEMS) based spatial light modulator (SLM) enclosed in a package filled with a fill gas having a low molar mass and high thermal conductivity to enhance the reliability and lifetime of the SLM is described. The MEMS based SLM can include one or more of a grating light valve (GLV™), a planar light valve (PLV™) or a linear PLV (LPLV), all of which are commercially available from Silicon Light Machines Inc., of San Jose CA, and are described in detail hereinafter. All of these devices are capable of phase and/or amplitude modulation with spatial filtering.

(7) Generally, the SLM includes a number of MEMS modulators, each including a number of light reflective surfaces, at least one light reflective surface coupled to an electrostatically deflectable element suspended above a substrate, and each adapted to reflect and modulate a light beam incident thereon. The package enclosing the SLM includes an optically transparent cover through which the reflective surfaces are exposed to the light beam, and a cavity is filled with a low molar mass fill gas having an atomic number of two or less and a thermal conductivity of greater than 100 milliwatts per meter kelvin (mW/(m.Math.K).

(8) In one embodiment, each of the MEMS based diffractors include a number of electrostatically deflectable ribbons suspended over a substrate, each ribbon having a light reflective surface.

Electrostatic deflection of one or more of the ribbons brings light reflected from the light reflective surface of a first electrostatically deflectable ribbon into interference with light reflected from the light reflective surface of ribbons in the diffractors in same or adjacent diffractors to modulate light incident thereon.
(9) In another embodiment, the MEMS based diffractors are two-dimensional diffractors, each including: a piston layer suspended over a surface of a substrate by posts at corners thereof, the piston layer including an electrostatically deflectable piston and a number of flexures through which the piston is coupled to the posts; a first reflective surface over a top surface of the piston; and a faceplate suspended over the piston layer, the faceplate including a second reflective surface on a top surface of the faceplate, and an aperture through which the piston is exposed. Electrostatic deflection of the piston brings light reflected from the first reflective surface into interference with light reflected from the second reflective surface to modulate light incident thereon.
(10) Further features and advantages of embodiments of the invention, as well as the structure and operation of various embodiments of the invention, are described in detail below with reference to the accompanying drawings. It is noted that the invention is not limited to the specific embodiments described herein. Such embodiments are presented herein for illustrative purposes only. Additional embodiments will be apparent to a person skilled in the relevant art(s) based on the teachings contained herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments of the present invention will be understood more fully from the detailed description that follows and from the accompanying drawings and the appended claims provided below, where:
(2) FIGS. **1**A and **1**B are schematic diagrams in a side and top view of an embodiment of a microelectromechanical systems (MEMS) based spatial light modulator (SLM) enclosed in a package filled with a fill gas having a low molar mass and high thermal conductivity;
(3) FIG. **1**C is a schematic block diagram of another embodiment of a MEMS based SLM enclosed in a package, and further including a system to pressurize the package with a fill gas having a low molar mass and high thermal conductivity;
(4) FIG. **1**D is a schematic block diagram of another embodiment of a MEMS based SLM enclosed in a flow through package, and further including a system to flow a fill gas having a low molar mass and high thermal conductivity through the package;
(5) FIGS. **2**A-**2**C are schematic block diagrams illustrating an embodiment of a MEMS based SLM including ribbon-type Microelectromechanical System modulators;
(6) FIGS. **3**A-**3**C are schematic block diagrams illustrating an embodiment of a MEMS based SLM including two-dimensional (2D) modulators;
(7) FIG. **4** is a schematic block diagram of a top view of a SLM including a multi-pixel, linear array of MEMS based 2D diffractors, such as those shown in FIGS. **3**A-**3**C;
(8) FIGS. **5**A-**5**C illustrate schematic diagrams of a 2D MEMS based complex SLM suitable for use with a phase-modulated system;
(9) FIGS. **6**A and **6**B show top views of portions of SLMs including a linear array of MEMS based modulators illustrating damage to the SLMs enclosed in cavities filled with a low molar mass and high thermal conductivity fill gas caused by increasing laser power, as compared to SLMs enclosed in nitrogen filled cavities;
(10) FIG. **7** shows top views of portions of SLMs including a linear array of MEMS based modulators enclosed in cavities filled with a low molar mass and high thermal conductivity fill gas illustrating the effect of fill gas pressure on damage caused by a laser operating at 14 kW/cm.sup.2

power;
(11) FIG. **8** is a graph illustrating relative thermal conductivity of a helium fill gas from 0.01 to 10 atmospheres for a MEMS-based modulators having different deflection gaps between the electrostatically deflectable element and a surface of a substrate;
(12) FIGS. **9**A-**9**C are perspective views of a single 2D modulator or PLV™ for a MEMS based SLM including large, thermal sinking structures filling previously void areas according to an embodiment of the present disclosure;
(13) FIG. **10** is a flow chart illustrating an embodiment of a method for operating a MEMS-based SLM packaged in a MEMS package filled with a low molar mass and high thermal conductivity fill gas;
(14) FIG. **11** is a block diagram of an optical system including a SLM including a linear array of MEMS based modulators enclosed in a package including a cavity filled with a low molar mass and high thermal conductivity fill gas;
(15) FIG. **12** is a schematic block diagram of a layout for a thermal, computer-to-plate (CtP) printing system including a SLM including a linear array of MEMS based modulators enclosed in a package including a cavity filled with a low molar mass and high thermal conductivity fill gas;
(16) FIG. **13** is a schematic block diagram of an additive manufacturing system including a SLM including a linear array of MEMS based modulators enclosed in a package including a cavity filled with a low molar mass and high thermal conductivity fill gas;
(17) FIG. **14** is a schematic block diagram of an embodiment of a surface modification system including a SLM including a linear array of MEMS based modulators enclosed in a cavity filled with a low molar mass and high thermal conductivity fill gas; and
(18) FIG. **15** is a schematic diagram of a Light Detection and Ranging (LiDAR) system including a SLM including a linear array of MEMS based modulators enclosed in a cavity filled with a low molar mass and high thermal conductivity fill gas.

DETAILED DESCRIPTION

(19) The present disclosure is directed to microelectromechanical systems (MEMS) based spatial light modulators (SLMs) enclosed in a MEMS package filled with a fill gas having a low molar mass and high thermal conductivity to enhance the reliability and lifetime of the SLM, and methods for operating the same in various applications are described.
(20) In the following description, numerous specific details are set forth, such as specific materials, dimensions and processes parameters etc. to provide a thorough understanding of the present invention. However, particular embodiments may be practiced without one or more of these specific details, or in combination with other known methods, materials, and apparatuses. In other instances, well-known semiconductor design and fabrication techniques have not been described in particular detail to avoid unnecessarily obscuring the present invention. Reference throughout this specification to “an embodiment” means that a particular feature, structure, material, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. Thus, the appearances of the phrase “in an embodiment” in various places throughout this specification are not necessarily referring to the same embodiment of the invention. Furthermore, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments.
(21) The terms “over,” “under,” “between,” and “on” as used herein refer to a relative position of one layer with respect to other layers. As such, for example, one layer deposited or disposed over or under another layer may be directly in contact with the other layer or may have one or more intervening layers. Moreover, one layer deposited or disposed between layers may be directly in contact with the layers or may have one or more intervening layers. In contrast, a first layer “on” a second layer is in contact with that second layer. Additionally, the relative position of one layer with respect to other layers is provided assuming operations deposit, modify and remove films relative to a starting substrate without consideration of the absolute orientation of the substrate.

(22) FIGS. **1**A and **1**B are schematic diagrams in a top and side view of an embodiment of a microelectromechanical system (MEMS) based spatial light modulator (SLM **100**). Referring to FIG. **1**A the SLM **100** generally includes a linear array **102** with a number of individual MEMS based light modulators **104** enclosed in a package **106** including a cavity **108** filled with a fill gas having a low molar mass and high thermal conductivity in which the linear array is enclosed, and an optically transparent cover **110** through which the number of modulators are exposed to a light beam. By low molar mass and high thermal conductivity it is meant an element having an atomic weight of two or less, i.e., helium (He) or hydrogen (H), and a thermal conductivity of greater than 100 milliwatts per meter kelvin (mW/(m.Math.K). Generally, each modulator **104** includes a number of light reflective surfaces (not shown in these figures) including at least one light reflective surface coupled to an electrostatically deflectable element to bring light reflected therefrom into constructive or destructive interference with another light reflective surface in the modulator or an adjacent modulator to modulate the amplitude or phase of a coherent light beam incident on the SLM **100**.
(23) Referring to FIG. **1**B, in the embodiment shown the package **106** is a surface-mount, no lead type of packaging, mechanically and electrically coupled to a circuit board **112** through a ball grid array (BGA **114**) or pin grid array (PGA) (not shown). The package **106** includes the cavity **108** in which the linear array **102** is enclosed, the optically transparent cover **110** through which reflective surfaces the modulators **104** are exposed, and wire bonds **116** to electrically couple the MEMS-based SLM to the BGA **114**. Generally, the linear array **102** is fabricated on a substrate **118** attached to the package **106** by a thin layer of solder **120**.
(24) In one embodiment, the cavity **108** is hermetically sealed and has a static pressure of at least 0.01 atmospheres. Preferably, the fill gas includes a gas mixture of from at least about 10 to about 100% hydrogen (H) or helium (He), at a static pressure of 0.1 atmospheres or greater, and more preferably, the fill gas has a pressure of about 1 bar or 1 atmosphere.
(25) Optionally or alternatively in another embodiment shown in FIG. **1**C the the MEMS package **106** further includes fill gas inlet or an inlet **122** by means of which the fill gas is introduced into the cavity **108** from a gas supply system **124**, to dynamically maintain a desired pressure of the fill gas therein. Generally, the gas supply system **124** includes a fill gas source **126**, such as a gas cylinder or Dewar storing a liquefied gas and a monitor, such as a mass flow controller (MFC **128**) to control and/or monitor the flow of fill gas to an inlet **122** of the package **106**, and a controller **130** to control operation of the MFC.
(26) In another embodiment shown in FIG. **1**D the package **106** is a flow through package further including a fill gas outlet or an outlet **132** opposite the inlet **122**, and the gas supply system **124** further includes a second MFC or solenoid valve **134**, through which the fill gas is exhausted or removed and is operable to provide a flow of gas from 0.01 standard cubic centimeters per minute (sccm) to 10000 sccm during operation of the SLM **100**.
(27) Generally, it has been found that the thermal conductivity and cooling of the MEMS based SLM is dependent, inter alia, on a thermal conductivity of the fill gas used. Table I below shows the thermal conductivity of various potential fill gases at 300 K.
(28) TABLE-US-00001 TABLE I Gas Thermal conductivity at 300 K (W/mK) Air 0.026 Ar 0.018 CO 0.025 CO.sub.2 0.017 H 0.182 He 0.151 N.sub.2 0.026 Ne 0.049 O.sub.2 0.027
(29) It is seen from table I that the thermal conductivity of Helium (He) is about 6 times that of N.sub.2. Thus, a thermal FEA (finite element analysis) predicts a 3.5 times improvement in cooling, and therefore power handling capability for the MEMS based SLM enclosed in a package including a cavity filled with helium (He) as compared to a cavity filled with air or nitrogen (N.sub.2). Furthermore, results of tests on a single 2D modulator of a MEMS based SLM, described in detail below, confirm that a surface of the modulators is illuminated by a 4 kW/cm.sup.2 photon flux and enclosed in a cavity filled with a nitrogen fill gas has temperature of ˜148° C. on a surface of the mirror, in contrast to a temperature of just 43° C. when a Helium fill

gas is used.
(30) An embodiment of a MEMS-based SLM including a multi-pixel, linear array of ribbon-type, electrostatically adjustable modulators, such as a GLV™, for which the package and method of the present disclosure is particularly useful will now be described with reference to FIGS. **2**A through **2**C. For purposes of clarity, many of the details of SLMs in general and MEMS-based ribbon-type modulators in particular that are not relevant to the present invention have been omitted from the following description. The drawings described are only schematic and are non-limiting. In the drawings, the size of some of the elements may be exaggerated and not drawn to scale for illustrative purposes. The dimensions and the relative dimensions may not correspond to actual reductions to practice of the invention.
(31) Referring to FIGS. **2**A and **2**B in the embodiment shown the SLM **200** includes a linear array **202** composed of thousands of free-standing, addressable electrostatically actuated ribbons **204**, each ribbon having a light reflective surface **206** supported over a surface of a substrate **208**, where a number of ribbons are grouped together to form the MEMS-based modulators. Each of the ribbons **204** includes an electrode **210** and is deflectable through a gap or cavity **212** toward the substrate **208** by electrostatic forces generated when a voltage is applied between the electrode **210** in the ribbon **204** and a base electrode **214** formed in or on the substrate. Each of the ribbon electrodes **210** are driven by one of a number of drive channels **216** in a driver **218**, which may be integrally formed on the same substrate **208** with the linear array **202**, as in the embodiment shown, or formed on a second substrate or chip and electrically coupled thereto (not shown).
(32) A schematic sectional side view of a ribbon **204** of the SLM **200** of FIG. **2**A is shown in FIG. **2**B. Referring to FIG. **1**B, the ribbon **204** includes an elastic mechanical layer **220** to support the ribbon above a surface **222** of the substrate **208**, a conducting layer or electrode **210** and a reflective layer **224** including the reflective surface **206** overlying the mechanical layer and conducting layer.
(33) Generally, the mechanical layer **220** comprises a taut silicon-nitride film, and is flexibly supported above the surface **222** of the substrate **208** by a number of posts or structures, typically also made of silicon-nitride, at both ends of the ribbon **204**. The conducting layer or electrode **210** can be formed over and in direct physical contact with the mechanical layer **220**, as shown, or underneath the mechanical layer. The conducting layer or ribbon electrode **210** can include any suitable conducting or semiconducting material compatible with standard MEMS fabrication technologies. For example, the conducting layer forming the electrode **210** can include a doped polycrystalline silicon (poly) layer, or a metal layer. Alternatively, if the reflective layer **224** is metallic it may also serve as the electrode **210**.
(34) The separate, discrete reflecting layer **224**, where included, can include any suitable metallic, dielectric or semiconducting material compatible with standard MEMS fabrication technologies, and capable of being patterned using standard lithographic techniques to form the reflective surface **206**.
(35) In the embodiment shown, a number of ribbons are grouped together to form a large number of MEMS channels or pixels **226**, each driven by a much smaller number of drive channels **216**. Deflection of a ribbon **204** causes light reflected from the reflective surface **206** to constructively or destructively interfere with light reflected from the reflective surface of an adjacent ribbon, there enabling the pixel **226** to switch between an on or bright state, an off or dark state or an intermediate gray-scale.
(36) Referring to FIG. **2**C in one embodiment the linear array **202** includes 1088 individually addressable ribbons **204** that can be grouped together to form channels or pixels **226** having any number of ribbons depending on pixel size requirements. Additionally, the SLM can include drive channels **216** (shown in FIG. **2**A) with up to 10-bit amplitude modulation to support gray-scale, and is capable of being modulated or switched at speeds up to 350 kHz. Referring again to FIG. **2**C, shaded rectangle illustrates an illuminated area **228** on the linear array **202** illuminated by a

rectangular beam directed onto the linear array **202** of the SLM **200**. In some embodiments for the SLM **200**, it is desirable to provide pixel configurations having a square aspect ratio. For example, in the embodiment shown wherein the linear array **202** includes thousands of ribbons **204**, each having a width of about 25 μm, and the illuminated area **228** has a width of about 75 μm, the ribbons can be grouped to form 360 square pixels **226***a* each including portions of three adjacent ribbons. Alternatively, the width of the illuminated area can be reduced to about 50 μm and the ribbons **204** can be grouped to form 512 50 μm×50 μm square pixels **226***b* each including portions of two adjacent ribbons, or the width of the illuminated area can be further reduced to about 25 μm such that each ribbon forms 1088 25 μm×25 μm square pixels **226***c*.

(37) Another type of SLM including a multi-pixel, linear array of MEMS-based two-dimensional (2D) modulators, such as a Linear Planar Light Valve (LPLV™) commercially available from Silicon Light Machines, Inc., of San Jose, California, for which the package and method of the present disclosure is particularly advantageous will now be described with reference to FIGS. **3**A through **3**C and FIG. **4**.

(38) For purposes of clarity, many of the details of fabricating and operating MEMS-based two-dimensional (2D) modulators, which are widely known and not relevant to the present invention, have been omitted from the following description. MEMS-based 2D modulators are described in greater detail, for example, in commonly assigned U.S. Pat. No. 7,064,883, entitled, “Two-Dimensional Spatial Light Modulator,” by Alexander Payne et al., issued on Jun. 20, 2006, and incorporated herein by reference in its entirety.

(39) FIG. **3**A illustrates a schematic block diagram of a sectional side view of a 2D diffractor or modulator **300** in a quiescent or un-driven state. Referring to FIG. **3**A, the 2D modulator **300** generally includes a piston layer **302** suspended over a surface of a substrate **304** by posts **306** at corners of the piston layer and/or 2D modulator. The piston layer **302** includes an electrostatically deflectable piston **302***a* and a number of flexures **302***b* through which the piston is flexibly or movably coupled to the posts **306**. A faceplate **308** overlying the piston layer **302** includes a first light reflective surface **310** and an aperture or cut-out portion **312** which separates the faceplate from a second reflective surface **314** on or attached to the piston **302***a*. The second light reflective surface **314** can either be formed directly on the top surface of the piston **302***a*, or, as in the embodiment shown, on a mirror **316** supported above and separated from the piston **302***a* by a central post **318** extending from the piston to the mirror. The first and second light reflective surfaces **310**, **314**, have equal area and reflectivity so that in operation electrostatic deflection of the piston **302***a* caused by an electrode **320** formed in or on the piston layer **302** and an electrode **322** in the substrate **304** brings light reflected from the first light reflective surface **310** into constructive or destructive interference with light reflected from the second light reflective surface **314**.

(40) Generally, the electrode **322** in the substrate **304** is coupled to one of a number drive channels in a drive circuit or driver **324**, which can be integrally formed in the substrate adjacent to or underlying the 2D modulator **300**, as in the embodiment shown. The electrode **322** in the substrate **304** can be coupled to the driver **324** through a via extending through the substrate from the driver to the electrode, and the electrode **320** formed in or on the piston layer **302** can be coupled to the driver or an electrical ground through a conductor extending through one of the posts **306** and the piston layer. As explained in greater detail below, typically multiple individual 2D modulators **300** are grouped or ganged together under control of a single drive channel to function as a single pixel in the multi-pixel, linear array of the SLM.

(41) FIG. **3**B is a schematic block diagram of the 2D modulator **300** of FIG. **3**A in an active or driven state, showing the piston **302***a* deflected towards the substrate **304**, and FIG. **3**C is a top view of the 2D modulator of FIGS. **3**A and **3**B illustrating the static, first light reflective surface **310** and the movable second light reflective surface **314**.

(42) An exemplary multi-pixel, linear array of dense-packed, MEMS-based 2D modulators will now be described with reference to the block diagram of FIG. **4**. FIG. **4** is a planar top view of a

SLM **400** including a linear array **401** of 2D modulators **402**, such as those shown in FIGS. **3**A-C, grouped or coupled together to a number of drive channels to or pixels.

(43) Referring to FIG. **4**, in one embodiment the 2D modulators **402** are grouped into a linear array **401** of interleaved channels or pixels **404** along a first, horizontal or longitudinal axis **406**. Each of the 2D modulators **402** in a single pixel **404** share a common drive channel or driver **408**. Although in the embodiment shown each pixel **404** is depicted as having a single column of 12 modulators **402** grouped along a transverse or vertical or transverse axis **410** perpendicular to the horizontal or longitudinal axis **406** of the array, this is merely to facilitate illustration of the array. It will be appreciated that each channel or pixel can include any number of 2D modulators arranged in one or more columns of any length across the width or vertical or transverse axis of the array without departing from the spirit and scope of the invention. For example, in one embodiment of the SLM **400** particularly suited for the spectral shaping systems and methods of the present disclosure, each pixel **404** includes a single column of 40 modulators grouped along the transverse axis **410** of the array. Similarly, the SLM **400** can include a linear array **401** having any number of pixels **404** or a number of individual linear arrays placed end to end adjacent to one another. This later configuration can help to increase power handling of the SLM **400** as the optically active area of the linear array **401** gets larger by increasing the number of columns of modulators per pixel. If the damage threshold per modulator is constant, power handling can be increased proportional to the area increase.

(44) In order to maximize or provide sufficient contrast for the SLM **400** it is desirable that incident light from an illumination source, have a numerical aperture (NA) or cone angle (Θ) which is smaller than the first-order diffraction angle (θ) of the diffractive SLM **400**. The diffraction angle (θ) of the SLM is defined as the angle between light reflected from a pixel **404** in the 0.sup.th order mode or state, and light reflected from the same pixel in the plus and/or minus 1.sup.st order mode. However, according to the grating equation, diffraction angles of a periodic surface, such as the linear array **401** of the SLM **400**, are set by a ratio of wavelength of light incident on the array to a spatial period or pitch of features of the periodic surface, i.e., the pixels **404**. In particular, the grating equation, equation 1 below, states:

$$\sin \theta = m\lambda/\Lambda \qquad (1)$$

where θ is a diffraction angle of light reflected from the surface, m is order of diffracted ray (integer), λ is the wavelength of the incident light, and Λ is a spatial or pitch of the modulator **402**. When we focus on a single pixel which has multiple modulators **402** and the incident light is ideal plane wave or has a numeric aperture (NA)=0, the light spreads due to Huygen-Fresnel principle. The spreading angle Θ is defined by equation 2 below as:

$$\Theta = \lambda/D \qquad (2)$$

where D is a pixel size.

(45) Achieving adequate contrast with conventional grating based SLMs requires either limiting illumination NA by means of an aperture (and suffering the associated throughput loss), or providing a large diffraction angle by reducing the size and spatial period or pitch of the individual modulators. However, this latter approach is problematic for a number of reasons including the need for larger, higher voltage drive circuits to drive smaller, movable grating elements, and a reduction of an optical power handling capability of the SLM resulting from such smaller grating elements.

(46) In contrast to conventional grating based SLMs, a SLM **400** including MEMS-based 2D modulators **402**, such as the LPLV™ is configured to have multiple pixels **404** each pixel including several modulators **402** arranged along the transverse or vertical axis **410** of the array (twelve in the embodiment shown), but with a much smaller number, generally only one or two modulators, arranged along the horizontal or longitudinal axis **406**. Because of this, the spreading angle Θ.sub.H of diffracted light from the pixel **404** along the longitudinal axis, where the pixel size is much smaller than along the vertical or transverse or transverse axis, is much larger than the

spreading angle Θ.sub.V of the pixel along the transverse axis. Conversely, the numerical aperture of illumination in the vertical direction (array short axis) can be much larger than the numerical aperture in the horizontal direction (array long axis) since the latter is limited by the diffraction angle of the SLM in order to achieve sufficient contrast. Thus by using a linear array of 2D modulators in combination with an asymmetric illumination NA in the longitudinal and transverse directions, the overall throughput of the spectral shaper can be improved.
(47) FIGS. **5**A-**5**C illustrate schematic diagrams of a 2D MEMS based complex SLM for phase-modulation, for which the package and method of the present disclosure is particularly useful. Briefly, the complex SLM **500** includes multiple pixels **502**, each pixel including multiple phase shift modulators **504**. The phase shift modulators **504** each include an electrostatically displaceable mirror or reflective surface **506**. The phase shift modulators **504** are configured such that substantially all light reflected from the complex SLM **500** comes from the phase shift modulators. Preferably, phase shift modulators **504** along diagonal lines **508**, **510**, are coupled to deflect in unison, by electrically interconnecting drive electrodes (not shown) below each phase shift modulator and applying a common drive voltage. In this way, each pixel **502** receives two independent driving voltages to deflect diagonally opposed phase shift modulators **504** as a group, denoted as group 1 and group 2 in FIG. **5**A. The two groups, of each pixel **502** can be controlled independently of the other pixels to allow coherent light reflected from one pixel to constructively or destructively interfere with light reflected from one or more adjacent pixels, thereby modulating the light incident thereon. More preferably, the phase shift modulators **504** are deflectable through one or more wavelengths of light to enable both the phase and the amplitude of the reflected light to be modulated independently. FIG. **5**B illustrates perspective views of a pixel **502** of the complex SLM **500** of FIG. **5**A in (a) quiescent, (b) phase-modulated and (c) amplitude and phase modulated mode, where δ is equal to a quarter wavelength of the light incident on the complex SLM.
(48) An exemplary embodiment of a phase shift modulator **504** of a complex SLM **500** will now be described in detail with reference to FIG. **5**C. FIG. **5**C depicts a detailed perspective view of a single phase shift modulator **504** in the complex SLM **500**. Referring to FIG. **5**C, the phase shift modulator **504** generally includes a film or membrane **512** disposed above an upper surface of a substrate **514** by a number of posts **516** with a displaceable or movable actuator or piston **518** formed therein. Supported above and affixed to each piston **518** by a support structure **520** is the reflective surface **506** that is positioned generally parallel to the surface of the substrate **514** and oriented to reflect light incident on a top surface of the complex SLM **500**. The piston **518** and the associated reflective surface **506** form an individual phase shift modulator **504**.
(49) Individual pistons **518** or groups of pistons are moved up or down over a very small distance (typically only a fraction of the wavelength of light) relative to the substrate **514** by electrostatic forces controlled by drive electrodes (not shown) in the substrate underlying the actuator membrane **512**. Preferably, the pistons **518** can be displaced by n*λ/2 wavelength, where λ is a particular wavelength of light incident on the complex SLM **500**, and n is an integer equal to or greater than 1. Moving the piston **518** brings reflected light from the planar light reflective surface **506** of one phase shift modulator **504** into constructive or destructive interference with light reflected by adjoining phase shift modulators in a pixel, thereby modulating light incident on the complex SLM **500**.
(50) FIGS. **6**A and **6**B show top views of portions of a SLM including a linear array of MEMS based modulators, such as shown in FIG. **4**, illustrating the effect of exposure to increasing laser power when enclosed in a cavity filled with nitrogen as compared to SLMs enclosed in cavities filled with a low molar mass and high thermal conductivity fill gas. Data for these illustrations were obtained using a 1064 nm laser with illumination optics operable to illuminate a 400 μm beam area of the linear array for a 5 second exposure at each of the different power levels. FIG. **6**A illustrates results of exposure at increasing laser power levels from 4 to 8 kW/cm2 on a linear array **600** enclosed in cavity filled with a nitrogen (N.sub.2) fill gas, at a pressure of about 31 pounds per

square in gauge (psig). FIG. **6**B illustrates results of exposure at increasing laser power levels from 12 to 20 kW/cm2 on a linear array **600** enclosed in cavity filled with a helium (He) fill gas, also at a pressure of about 31 psig. Referring to FIGS. **6**A and **6**B it is seen that these exposure tests (repeated twice for each fill gas) show a 2.5 to 3 times improvement in power handling capability for a linear array **600** enclosed in a He filled cavity over one filled with N.sub.2. That is the power in kW/cm.sup.2 that can be applied before damage to the SLM is about 2.5 to 3 times greater. In particular it is noted that the linear array **600** enclosed in the He filled cavity was exposed to a laser power of about 14 kW/cm.sup.2 before damage was incurred similar to that seen for the linear array **600** enclosed in a nitrogen (N.sub.2) filled cavity and exposed to a laser power of about 5 kW/cm.sup.2.
(51) FIG. **7** shows top views of portions of a SLM including a linear array **700** of MEMS based modulators illustrating the ability of different pressures of a low molar mass and high thermal conductivity fill gas increase thermal conduction, thereby decreasing damage to the SLM, and increasing power handling capability while increasing reliability and longevity of the SLM. Data for these illustrations were obtained using a 1064 nm laser with a beam area of 400 μm operating at a power of about 14 kW/cm.sup.2 for a 5 second exposure at each of the different pressures from 5 to 35 psig. Referring to FIG. **7** it is seen that a strong dependence was found, with damage reduced with increasing pressure, and the damage greatly between 10 and 15 psig, at atmospheric pressure of about 1 atmosphere or 14 psig.
(52) FIG. **8** is a graph illustrating relative thermal conductivity of a helium fill gas from 0.01 to 10 atmospheres for MEMS-based SLM having different deflection gaps between the electrostatically deflectable element and a surface of a substrate. FIG. **8** was derived by calculating the rarified gas thermal conductivity equation 3 below, where the terms of the equation are given in Table II.
(53) $\frac{K}{K_c} = \frac{1}{1 + C\frac{T}{pd}}$ (3)
(54) TABLE-US-00002 TABLE II C (mK/N) constant equal to 7.65E−5 T (K) absolute temperature in Kelvin p (Pa) Pressure in bar or atmospheres d (m) deflection gap K (W/mK) thermal conductivity K.sub.0 (W/mK) thermal conductivity at 1 bar K/K.sub.0 — thermal conductivity ratio
(55) Referring to FIG. **8** it is noted that as dimensions of the MEMS-based SLM, i.e., the deflection gap, are reduced thermal conductivity becomes a function of a pressure of the fill gas. By rarified it is meant a pressure less than 1 bar or atmosphere. In FIG. **8** line **802** represents the thermal conductivity of a helium fill gas from 0.01 to 10 atmospheres for a MEMS-based SLM a 1 mm deflection gap; line **804** represents the thermal conductivity for He fill gas with a MEMS-based SLM having a 100 μm deflection gap; line **806** represents the thermal conductivity for He fill gas with a MEMS-based SLM having a 10 μm deflection gap; line **808** represents the thermal conductivity for He fill gas with a MEMS-based SLM having a 1 μm deflection gap; and line **810** represents the thermal conductivity for He fill gas with a MEMS-based SLM having a 100 nm deflection gap. Referring to FIG. **8** it is seen that at reduced dimensions of a deflection gap between an electrostatic movable element, such as piston layer **302** and a surface of a substrate **304**, thermal conductivity varies primarily as a function of fill gas pressure. It is further noted that for a MEMS based SLM, such as a linear PLV™ or LPLV described above, the results of the tests shown graphically that at pressures below 1 atmosphere the fill gas behaves as a rarified gas in which thermal conductivity drops and becomes a function of pressure. By rarified it is meant a pressure less than 1 bar or atmosphere.
(56) In another embodiment, the thermal conductivity or transfer of heat away from a MEMS-based SLM can be increased, and the reliability and lifetime of the SLM enhanced by the inclusion of thermal sinking structures in the modulators. FIGS. **9**A-**9**C are perspective views of a lower MEMS or piston layer of a PLV™ modulator including large, thermal sinking structures filling previously void areas according to an embodiment of the present disclosure. In particular, FIG. **9**A

is a perspective view of a first MEMS or piston layer **902** of an incomplete PLV™ **900** suspended above a substrate **904** by a number of posts **906** in corners thereof, and including large, thermal sinking structures **908** substantially filling previously void areas in the piston layer to provide thermal management of the piston layer. The piston layer **902** includes an electrostatically deflectable piston **902***a* and a number of flexures **902***b* through which the piston is flexibly or movably coupled to the posts **906**. The posts **906** can include an annular structure including an outer or first material **906***a* surrounding either a substantially hollow inner region or, as shown, a second material **906***b*. The first and second materials of the posts **906** can include a dielectric, conductive, or semiconductor material, selected to be compatible with the material and processes used to form the PLV™. Generally, at least one of the first or second materials of at least one of the posts **906** includes a conductive or semiconductor material to electrically couple an electrode (not shown in this figure) in or on the piston **902***a* to integrated drive circuitry (not shown in this figure) formed in or electrically coupled to the substrate **904**.

(57) In the embodiment shown in FIG. **9**A the thermal sinking structures **908** include ends or upper surfaces substantially co-planar with an upper surface of the piston layer **902**, and which do not extend substantially through or past the piston. The thermal sinking structures **908** can include a solid, homogeneous structure or an annular structure including an outer or first material **908***a* surrounding either a substantially hollow inner region or a second material **908***b*, as shown in FIG. **9**D. The first and second materials of the thermal sinking structures **908** can include a dielectric, conductive, or semiconductor material. In one embodiment, the thermal sinking structures **908** include SiGe surrounding a core of Ge. An outer or first material **908***a* including SiGe is desirable as providing resistance to etchants used in forming the PLV™, while also providing a conductive path to electrically couple an electrode in or on the piston **902***a* to integrated drive circuitry in or electrically coupled to the substrate **904**. The annular structure of the thermal sinking structures **908** is desirable as enabling the thermal sinking structures to have a large outer surface area, substantially filling void spaces between the posts **906**, the flexures **902***b* and the piston **902***a* of the piston layer **902**, without requiring excessive deposition of a conformal layer of the first material of the of the thermal sinking structures. As explained in further detail below this is particularly advantageous when the first material of the first material **908***a* of the thermal sinking structures **908** is deposited in a single step concurrent with, for example forming the piston layer **902** and or the posts **906**. In some embodiment, such as that shown in FIG. **9**D, the inner core or region of the thermal sinking structures **908** is not hollow, but is filled with a second material **908***b*, such a germanium (Ge) having an greater thermal conductivity than the first material **908***a*. Silicon-germanium (SiGe) has a thermal conductivity, depending on the ratio of silicon to germanium of from about 0.089 W/cm-° C. to about 0.11 W/cm-° C., while germanium (Ge) has a thermal conductivity of about 0.6 W/cm-° C. As explained in further detail below germanium (Ge) is also suitable as a sacrificial material useful in forming the PLV™, thus eliminating the need for a separate deposition step to deposit the second material **908***b*. It is further noted that use of Ge as a sacrificial material also enables easy replacement of void while easily maintaining planarity of a second sacrificial layer.

(58) FIG. **9**B is a perspective view of a complete PLV™ similar in structure to that of FIG. **9**A and further including a second MEMS or faceplate layer **910** including a reflective top surface suspended above the piston layer **902** by second posts **912** and a mirror **914** formed either on or above and attached to the piston **902***a* of the piston layer. As with the first posts **906**, the second posts **912** can include an annular structure including an outer or first material surrounding either a substantially hollow inner region or, as shown, a second material. As in PLVs™ described above, in the embodiment shown in FIG. **9**B the mirror is substantially co-planar with an upper surface of the faceplate layer **910** when the PLV™ is in a quiescent or unpowered state and the piston **902***a* is not deflected. However, it will be understood that this placement of the mirror **914** in the state can be changed without changing or adversely impacting the structures and methods of the present

invention. In particular, it is noted that mirror **914** can be positioned above or below the reflective top surface of the faceplate layer **910** by an even or odd multiple of one quarter (¼) wavelength of light modulated by the PLV™ to provide either constructive or destructive interference with light reflected from the reflective top surface of the faceplate layer and/or to modulate a phase of the reflected light. In the embodiment shown in FIG. **9**A the thermal sinking structures **908** include ends or upper surfaces which terminate proximal to the faceplate layer **910** to provide thermal management of the faceplate layer as well as improved thermal management of the piston layer of the PLV™.

(59) FIG. **9**C is a see through, perspective view of the PLV™ including a faceplate layer **910** suspended above the piston layer **902** (shown in phantom), suspended above the surface of the substrate **904** by posts **912** in four corners of the PLV™, and including large, thermal sinking structures **908** extending through or past the piston layer to terminate proximal to or in contact with a faceplate. The thermal sinking structures **908** can include an annular-ring shape comprising first and second materials. It is further noted that this design, i.e., a post having an annular-ring shaped cross-section, can also be applied to the posts **906**, **912** at the four corners of the PLV™.

(60) FIG. **9**C also illustrates a central post **916** (shown in phantom) supporting the mirror **914** above the piston layer **902**, and, optionally, to provide thermal management of the mirror. In the embodiment, shown the central post **916** has an annular-ring shaped cross-section including an annular-ring of first material **916***a* surrounding either a substantially hollow inner region or an inner core of a second material **916***b*. As with the thermal sinking structures **908**, the first and second materials of the central post **916** can include a dielectric, conductive, or semiconductor material. In one embodiment, the central post **916** includes SiGe surrounding a core of Ge. Previously, it was not possible enlarge the size of the posts **906**, **912**, without creating dimples on a surface of the faceplate layer **910**, mirror **914** or piston layer **902**. Thus, sizes of the posts **906**, **912**, **916** were limited by a thickness of the piston or mirror and faceplate layers in order to provide a smooth, substantially planar top surface.

(61) A method for operating a MEMS-based SLM packaged in a MEMS package filled with a low molar mass and high thermal conductivity fill gas will now be described with reference to the flow chart of FIG. **10**. Referring to FIG. **10**, the method begins with providing a MEMS based SLM including reflective surfaces enclosed within a cavity of a package including an optically transparent cover (step **1002**). Next, the cavity is filled with a low molar mass fill gas having an atomic number of two or less and a thermal conductivity of greater than 100 mW/(m.Math.K), at a pressure of 0.1 atmosphere or greater (step **1004**). In some embodiments, the cavity is filled with a low molar mass fill gas having a thermal conductivity of 151 mW/(m.Math.K), at a pressure of 1 atmosphere. Optionally, a fluid flow is established the through cavity from an inlet and an outlet in the package (step **1006**). The reflective surfaces are then exposed to a light beam through the optically transparent cover (step **1008**). Again, optionally the method can further include monitoring pressure in the cavity during exposure of the reflective surface to the light beam (step **1010**), and stopping exposure of the reflective surface to the light beam if the pressure drops below a specified minimum 0.1 atmospheres (step **1012**).

(62) FIG. **11** is a block diagram of an optical system **1100** including a SLM **1102** with a multi-pixel, linear array of MEMS-based modulators (not shown in this figure), such as shown in FIGS. **2**A through **2**C, or a multi-pixel, linear array of 2D modulators, such as shown in FIGS. **3**A through **3**C and FIG. **4**. Briefly, the optical system **1100** includes, in addition to the SLM **1102**, a light source, such as a laser **1104**, operable to illuminate the SLM, imaging optics **1106** operable to focus a substantially linear swath of modulated light onto an imaging or focal plane, such as surface of a workpiece on a fixture **1108**, and a controller **1110** operable to control the SLM, laser and imaging optics to scan the linear swath of modulated light across the surface of the workpiece to record a two-dimensional (2D) image thereon. Generally, as in the embodiment shown, the optical system **1100** further includes illumination optics **1112** with a beam forming optical system to direct a

rectangular beam onto the linear array of the SLM **1102**.

(63) Typically, the laser **1104** is capable of operating in ultraviolet (UV) wavelengths of from 355 nanometers (nm) through infrared (IR) wavelengths up to about 2000 nm in either a continuous wave (CW) mode, or in a pulse mode with widths or durations of from about 1 femtoseconds (fs) up to about 500 nanoseconds (ns) at a repetition rate of from about 10 kHz up to about 300 kHz, and at energy ranges of from about 10 microjoules (μJ) up to greater than 10 millijoules (mJ).

(64) The imaging optics **1106** can include dynamic optical elements, such as galvanometric mirrors, to scan the linear swath of modulated light across the surface of the workpiece, and a number of static optical elements to direct modulated light to the galvanometric mirrors and/or to focus the modulated light from the galvanometric mirrors onto the surface of the workpiece.

(65) The fixture **1108** on which the workpiece is placed or affixed can include a static fixture, or a movable stage operable to move or reposition the workpiece relative to a substantially stationary linear swath of modulated light, to scan the linear swath of modulated light across the surface of the workpiece. In either embodiment, whether static or movable, the fixture **1108** preferably includes a number of sensors and signaling means to signal other components in the laser marking system when the workpiece is in proper position to be marked. In some embodiments, the fixture **1108** includes a movable stage capable of being moved along two orthogonal axes to enable scanning multiple parallel swaths to scan larger areas on the workpiece. In other embodiments, the optical system **1100** includes both imaging optics **1106** with galvanometric mirrors, and a movable stage (fixture **1108**) capable of being moved along a single axis orthogonal to the direction the galvanometric mirrors scan the linear swath of modulated light to scan larger areas on the workpiece.

(66) The laser **1104**, illumination optics **1112**, SLM **1102**, imaging optics **1106** and workpiece held on the fixture **1108** are optically coupled in the direction indicated by arrows **1114**. Additionally, the laser **1104**, illumination optics **1112**, SLM **1102**, imaging optics **1106** and fixture **1108** are electrically coupled in signal communication with the controller **1110** and each other through a control bus **1116**. In particular, controller **1110** provides digital image data to the SLM **1102**, controls a power level of the laser **1104**, controls operation of galvanometric mirrors in the imaging optics **1106** and controls the movable stage of the fixture **1108** (where included) through the control bus **1116**. Additionally, the fixture **1108** can signal the controller **1110**, SLM **1102** and/or the laser **1104** when the workpiece is in proper position, and the SLM can signal the laser when the image data loaded to the SLM is ready to be scanned on the workpiece so that the laser can be pulsed.

(67) FIG. **12** is a schematic block diagram of a layout for a thermal, computer-to-plate (CtP) printing system including a SLM having a linear array of MEMS based modulators enclosed in a package including a cavity filled with a low molar mass and high thermal conductivity fill gas. Referring to FIG. **12**, a printing system or printer **1200** including a SLM **1202** having a linear array of MEMS based modulators, such as shown in FIGS. **2**A through **2**C, or a multi-pixel, linear array of 2D modulators, such as shown in FIGS. **3**A through **3**C and FIG. **4**, is shown in FIG. **12**. An advantage of using the SLM **1202** having a linear array of MEMS based modulators in this application is that it eliminates the need for a scanning mirror and f-O or scanning optics of conventional SLM by replacing them with the SLM **1202** having a linear array of MEMS based modulators sufficiently large along a longitudinal axis to project modulated light over a swath extending substantially across the entire width of an imaging or focal plane. The printer **1200** generally comprises an illuminator or light source **1204**, illumination optics **1206**, and imaging optics **1208** to direct an image (modulated light) from the SLM **1202** onto a photosensitive or photoconductive surface of the imaging or focal plane, shown here as a drum **1210** with a photoconductive layer on a surface **1212** thereof.

(68) The light source **1204** can include any light emitting device capable of continuously emitting light at a sufficient power level or power density, and, preferably at a single or narrow range of wavelengths to enable light reflected from modulators of the SLM **1202** to be modulated in phase

and/or amplitude by diffraction. In certain printing applications, and in particular in photothermal printers, the light source **1204** can include a number of lasers or laser emitters, such as diode lasers, each powered from a common power supply (not shown) in a CW (Continuous Wave) operation. Preferably, the light source **1204** is a high-power diode laser producing from about 5000 to about 40,000 milliwatts (mW) of power at a wavelength (λ) of from about 750 to about 1000 nm.
(69) The illumination optics **1206** can comprise a number of elements including lens integrators, mirrors and prisms, designed to transfer light from the light source **1204** to the SLM **1202** such that a line of a specified length and width is illuminated on the SLM. In the embodiment shown, the illumination optics **1206** include a prism **1214** and lens **1216** to refract and transmit light from the light source **1204**, and an integrator **1218** to illuminate a swath covering substantially the full width of the SLM.
(70) The imaging optics **1208** can include magnification elements, such as a Fourier Transform (FT) lens **1220** and a FT mirror **1222**, to direct the light from the SLM **1202** to the photoconductive layer located on the drum **1210**. Preferably, the imaging optics **1208** is designed to transfer light from the SLM **1202** to the drum **1210** such that a photoconductive layer located on the drum is illuminated across a swath covering substantially the full width of the drum. Optionally, as in the embodiment shown, the imaging optics **1208** further includes filter elements, such as a FT filter **1224**, to resolve light reflected from each pixel but not light reflected from each individual modulator or diffractor or from each element in each modulator.
(71) As described above with reference to FIGS. **3**A-C and **4**, the SLM **1202** includes a linear array of a number of individual diffractive two-dimensional (2D) densely packed spatial light modulators or diffractors (not shown in this figure). Adjacent 2D modulators may be grouped or functionally linked to provide a number of pixels in the linear array that can be controlled by drive signals from a single, common drive channel to print to the imaging or focal plane with a desired resolution. Preferably, the SLM **1202** has a pixel count adequate to cover a swath or strip extending substantially across the entire width of the photo or thermal-sensitive surface of the imaging or focal plane. More preferably, the SLM **1202** has a pixel count of at least about 500 pixels, and most preferably of at least about 1000 pixels to provide the desired resolution. For example, in one version of the layout illustrated in FIG. **12**, the SLM **1202** includes a sufficient number of pixels to cover an entire swath on a standard eight inch (8″) write drum **1210** with a printing resolution of about 2000 dots-per-inch (dpi) using a modest-power, 780 nm GaAs diode laser as the light source **1204**.
(72) In another embodiment, not shown, the printing system is a maskless lithography or photolithography system including the SLM and further comprising a pattern generator to generate and transmit to the number of 2D modulators drive signals to manufacture micro-electronic devices. By micro-electronic devices it is meant integrated circuits (ICs) and Micro-Electromechanical System or MEMS devices. In maskless lithography, light used to expose the photosensitive material in an image plane on a substrate, such as silicon or semiconductor wafer, on which the device is to be formed.
(73) FIG. **13** is a schematic block diagram of a three-dimensional (3D) printing or additive manufacturing system including a SLM enclosed in a package including a cavity filled with a low molar mass and high thermal conductivity fill gas. Referring to FIG. **13**, the 3D printing system **1300** generally includes a MEMS-based diffractive SLM **1302** with a number of pixels, each with multiple modulators to modulate a beam of light generated by a laser **1304**, a vat **1306** containing the photopolymer or resin **1308** (indicated by dashed lines), and a transport mechanism **1310** to raise and lower a work surface **1312** on which an object **1314** is printed into the vat as indicated by the vertical arrow. The laser **1304** is capable of operating in UV wavelengths (λ) of from 355 nm through IR wavelengths up to about 2000 nm in either CW mode, or in a pulse mode with widths or durations of from about 1 fs up to about 500 ns at a repetition rate of from about 10 kHz up to about 300 kHz, and at energy ranges of from about 10 microjoules (μJ) up to greater than 10

millijoules (mJ).

(74) Referring again to FIG. **13**, the 3D printing system **1300** further includes imaging optics **1318** to transfer modulated light from the SLM toward the work surface **1312**, a controller **1316** to control operation of the laser **1304**, SLM **1302** and transport mechanism **1310**. In some embodiments, the imaging optics **1318** can include magnification and filtering elements, such as a first Fourier Transform (FT) lens to focus and direct light from the SLM **1302** onto a scanning mirror **1320** that rotates to scan a modulated beam **1322** along a x-axis and a y-axis on a surface of the resin **1308** immediately above or adjacent to the work surface **1312**.

(75) The transport mechanism **1310** is adapted and controlled by the controller **1316** to lower the work surface **1312** into the vat **1306** as the modulated light converts the liquid resin **1308** into a solid, building successive layers or cross-sections of the object **1314** to be printed. Generally, the 3D printing system **1300** further includes a sweeper **1324** adapted to move as indicated by the horizontal arrow to spread or smooth fresh resin **1308** over surface sections of the object **1314** being printed.

(76) FIG. **14** is a schematic block diagram of an embodiment of a laser marking system including a SLM enclosed in a cavity filled with a low molar mass and high thermal conductivity fill gas. FIG. **14** is a schematic block diagram of an embodiment of a laser marking system **1400** including a SLM **1402** with a multi-pixel, linear array of MEMS based modulators, and galvanometric mirrors for scanning. For purposes of clarity and to simplify the drawings the optical light path is shown as being unfolded causing the SLM **1402** to appear as transmissive. However, it will be understood that because the SLM **1402** is reflective the actual light path is folded to an angle of 90° or less at the SLM.

(77) Referring to FIG. **14**, the laser marking system **1400** includes, in addition to the SLM **1402**, a laser **1404** operable to generate laser light used to illuminate the SLM, illumination optics **1406** to direct laser light onto the SLM, imaging optics **1408** operable to focus a substantially linear swath of modulated light **1410** onto a surface **1412** of a workpiece **1414** on or affixed to a fixture **1416** or stage, and a controller **1418** operable to control the SLM, laser and imaging optics to scan the linear swath of modulated light across the surface of the workpiece to record a 2D image thereon. The laser **1404** is capable of operating in UV wavelengths (λ) of from 355 nm through IR wavelengths up to about 2000 nm in either CW mode, or in a pulse mode with widths or durations of from about 1 fs up to about 500 ns at a repetition rate of from about 10 kHz up to about 300 kHz, and at energy ranges of from about 10 microjoules (μJ) up to greater than 10 millijoules (mJ).

(78) The SLM **1402** can include a multi-pixel; linear array of MEMS based, ribbon-type modulators, such as shown in FIGS. **2**A through **2**C, or a multi-pixel, linear array of 2D modulators, such as shown in FIGS. **3**A through **3**C and FIG. **4**.

(79) The illumination optics **1406** can include a beam forming optical system to direct laser light onto the SLM **1402**. Referring to FIG. **14**, elements of the beam forming optical system can include a Powell lens **1420**, a long axis collimating lens **1421**, and a cylindrical, short axis focusing lens **1422** to shape or focus the illumination into a rectangular beam or line of illumination extending substantially uniformly across the linear array of the SLM **1402**.

(80) The imaging optics **1408** can include galvanometric mirrors **1424** to scan the linear swath of modulated light **1410** across the surface **1412** of the workpiece **1414**, a number of cylindrical lens **1426** to direct modulated light to the galvanometric mirrors, and a Fourier aperture **1428** to separate a 0.sup.th order beam in the modulated light from 1.sup.st order beams, and a Fourier Transform (FT) lens **1430** to focus the modulated light onto the surface of the workpiece.

(81) Preferably, the cylindrical lens **1426** and FT lens **1430** of the imaging optics include fused silica lenses to reduce thermal focus shift of the modulated light focused onto the surface **1412** of a workpiece **1414**. In some embodiments, one or more of the lenses **1420**, **1421**, **1422** of the illumination optics **1406** can also include fused silica lenses to reduce thermal focus shift of the laser light focused onto the SLM **1402**.

(82) The fixture **1416** on which the workpiece **1414** to be marked is placed or affixed can include a static fixture, or a movable stage operable to move or reposition the workpiece relative to a substantially stationary linear swath of modulated light, to scan the linear swath of modulated light across the surface of the workpiece. As noted above, in either embodiment, whether static or movable, the fixture **1416** preferably includes a number of sensors and signaling means to signal other components in the laser marking system when the workpiece is in proper position to be marked.

(83) Depending on the size of the linear swath of modulated light **1410** and/or an image to be recorded it can be recorded on the surface **1412** of a workpiece **1414** in a single scan or single-stripe of the linear swath of modulated light **1410** across the surface along a single axis, or by multiple scans or stripes (multi-stripes) of the linear swath of modulated light across the surface along a first axis perpendicular to a long axis of the linear swath of modulated light, followed by repositioning the linear swath of modulated light along a second axis parallel to the long axis of the linear swath.

(84) FIG. **15**A is a schematic diagram of a Light Detection and Ranging (LiDAR) system including a SLM enclosed in a cavity filled with a low molar mass and high thermal conductivity fill gas. FIG. **15** is a schematic functional diagram of a portion of a LiDAR system **1500** including controller **1501** and a solid state optical scanner **1502** having an optical transmitter **1504** with at least one MEMS phased-array **1506** enclosed in a cavity filled with a low molar mass and high thermal conductivity fill gas, and configured to receive light from a light source through shaping or illumination optics (not shown in this figure), and to modulate phases of at least some of the received light and transmit or project a beam of phase modulated light through projection optics **1508** operable to project and steer a line or swath **1510** of illumination to scan a far field scene **1512**. The MEMS phased-array **1506** steers the beam of light to scan the far field scene **1512** by changing phase modulation of light incident on different portions of the MEMS phased-array. Generally, the first MEMS phased-array **1506** is configured to scan the far field scene **1512** in at least two-dimensions (2D), including an angular dimension (θ), and an axial dimension (indicated by arrow **1514**) parallel to a long axis of the MEMS phased-array.

(85) It is noted that although the optical scanner **1504** is shown schematically as including a single MEMS phased-array **1506** this need not and generally is not the case in every embodiment. Rather, as explained in detail below, it is often advantageous that the optical scanner **1504** include multiple adjacent MEMS phased-arrays **1506** operated in unison or a single MEMS phased-array having multiple adjacent arrays to increase either aperture width or length to increase a power or radiant flux of the transmitted or received light and to increase the point spread resolution of the system.

(86) The optical scanner **1504** further includes an optical receiver **1516** including collection or receiving optics **1518** to capture light from the far field scene **1512**, which is then directed onto a detector **1520**.

(87) Referring to FIG. **15**, depth or distance information from the LiDAR system **1500** to a target or object **1522** in the far field scene **1512** can be obtained using any one of a number of standard LiDAR techniques, including pulsed, amplitude-modulated-continuous-wave (AMCW) or frequency-modulated-continuous-wave (FMCW). In pulsed and AMCW LiDAR systems the amplitude of intensity of the light transmitted is either pulsed or modulated with a signal, and the TOF from the LiDAR system **1500** to the object **1522** is obtained by measuring the amount by which the return signal is time-delayed. The distance to the reflected object is found by multiplying half this time by the speed of light.

(88) FIG. **15**B is a diagram illustrating a change in frequency of an outgoing pulse of transmitted light over time for a LiDAR system using an FMCW technique. Referring to FIGS. **15**A and **15**B, in a FMCW LiDAR the frequency of an outgoing chirp or pulse **1524** of transmitted light is continuously varied over time as the swath **1510** of light is continuously scanned across the far field scene **1512**. The time to the object **1522** can be determined by comparing the frequency of

light reflected from the object to that of a local oscillator, and the distance to the object is found by using the speed of light as previously described. FMCW LiDAR systems have an advantage over amplitude modulated in that the local oscillator provides an inherent amplification of the detected signal.

(89) With information on TOF the controller **1501** in the LiDAR system can then calculate a location of the object **1522** in the far field scene **1512** along an X-axis **1526** from the steering direction of the MEMS phased-array **1506** when the light was transmitted from the MEMS phased-array, and along a Y-axis **1528** from a sensed location of the object along an axis of the detector **1520** (indicated by arrow **1530**) parallel to a long axis of the detector.

(90) Thus, embodiments of a MEMS based SLMs enclosed in helium filled MEMS packages, and methods for fabricating and operating the same in various applications have been described. Although the present disclosure has been described with reference to specific exemplary embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader spirit and scope of the disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

(91) The Abstract of the Disclosure is provided to comply with 37 C.F.R. § 1.72(b), requiring an abstract that will allow the reader to quickly ascertain the nature of one or more embodiments of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

(92) Reference in the description to one embodiment or an embodiment means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the circuit or method. The appearances of the phrase one embodiment in various places in the specification do not necessarily all refer to the same embodiment.

Claims

1. A spatial light modulator (SLM) comprising: a number of Microelectromechanical System (MEMS) modulators, each modulator comprising a number of light reflective surfaces including at least one light reflective surface coupled to an electrostatically deflectable element suspended above a surface of a substrate, and each modulator adapted to reflect and modulate a light beam incident thereon; and a package enclosing the number of MEMS modulators, the package comprising a cavity in which the number of MEMS modulators are enclosed, an optically transparent cover through which the number of reflective surfaces are exposed to the light beam, and an inlet to the cavity, wherein the inlet is coupled to a gas supply and is operable to provide in the cavity a low molar mass fill gas having an atomic number of two or less and a thermal conductivity of greater than 100 milliwatts per meter kelvin (mW/(m.Math.K), and to maintain a pressure of 0.1 atmospheres or greater when the SLM is operated to modulate the light beam.
2. The SLM of claim 1 wherein the fill gas comprises from 10 to 100% hydrogen (H) or helium (He).
3. The SLM of claim 2 wherein the fill gas comprises a mixture of hydrogen (H) or helium (He).
4. The SLM of claim 1 wherein the package further comprises an outlet operable to provide a flow of gas of from 0.01 Standard Cubic Centimeters per Minute (sccm) to 10000 sccm during operation of the SLM.
5. The SLM of claim 1 wherein each of the number of MEMS modulators comprises a number of

electrostatically deflectable ribbons suspended over the surface of the substrate, each ribbon having a light reflective surface, wherein electrostatic deflection of the number of electrostatically deflectable ribbons brings light reflected from the light reflective surface of a first electrostatically deflectable ribbon into interference with light reflected from the light reflective surface of a second electrostatically deflectable ribbon.

6. The SLM of claim 1 wherein each of the number of MEMS modulators comprises a two-dimensional (2D) MEMS modulator including: a piston layer suspended over the surface of the substrate by posts at corners thereof, the piston layer including an electrostatically deflectable piston and a number of flexures through which the piston is coupled to the posts; a first reflective surface over a top surface of the piston; and a faceplate suspended over the piston layer, the faceplate including a second reflective surface on a top surface of the faceplate, and an aperture through which the piston exposed, wherein electrostatic deflection of the piston brings light reflected from the first reflective surface into interference with light reflected from the second reflective surface.

7. The SLM of claim 6 wherein the 2D MEMS modulators are arranged as a linear array.

8. The SLM of claim 1 wherein the number of MEMS modulators are operable to modulate one or both of a phase and an amplitude of light incident thereon.

9. A system comprising: a light source operable to generate a light beam; a spatial light modulator (SLM) operable to modulate light incident thereon, the SLM including a multi-pixel, linear array of MEMS modulators, the SLM enclosed in a cavity in a package with an optically transparent cover through which the linear array of MEMS modulators are exposed to the light beam, the cavity filled a fill gas comprising from 10 to 100% hydrogen (H) or helium (He); imaging optics operable to focus modulated light from the SLM onto a focal plane of the system; a controller operable to control the light source, the SLM, and the imaging optics to scan the modulated light across the focal plane, and a fill gas source coupled to a fill gas inlet on the package of the SLM to provide fill gas to the cavity, and wherein the controller is further operable to control the fill gas source to maintain a pressure of 0.1 atmospheres or greater during operation of the SLM.

10. The system of claim 9 further comprising fill gas outlet on the package of the SLM, and wherein the controller is further operable to control the fill gas source to provide a flow of gas of from 0.01 Standard Cubic Centimeters per Minute (sccm) to 10000 sccm during operation of the SLM.

11. The system of claim 9, wherein the system is an additive manufacturing system further comprising: a vat into which material being added together is introduced; and a transport mechanism to raise and lower a work surface, on which an object is to be manufactured into the vat, wherein the focal plane is on a surface of the material in the vat, and wherein the controller is further operable to control operation of the transport mechanism.

12. The system of claim 9, wherein the system is a surface modification system, the focal plane is on a surface of a workpiece, the light source is a laser, and wherein the imaging optics is operable to focus a substantially linear swath of modulated light onto the surface of the workpiece, the linear swath comprising light from multiple pixels of the SLM, and wherein the controller is operable to control the SLM, laser and imaging optics to mark the surface of the workpiece to record an image thereon.

13. The system of claim 9, wherein the system is a light detection and ranging (LiDAR) system, the focal plane is far field scene, the light source is a coherent light source, the imaging optics are operable to project light from the SLM to the far field scene, and further comprising receiving optics to receive light from the far field scene and direct received light onto a detector.

14. A method comprising: providing a Micro-Electromechanical System (MEMS) based spatial light modulator enclosed within a cavity of a package comprising an optically transparent cover through which reflective surfaces of the MEMS based SLM are exposed, and an inlet to the cavity; and coupling the inlet to a fill gas source and filling the cavity with a low molar mass fill gas

having an atomic number of two or less and a thermal conductivity of greater than 100 milliwatts per meter kelvin (mW/(m.Math.K); operating the fill gas source to maintain a pressure of 0.1 atmospheres or greater during operation of the MEMS based spatial light modulator; and exposing the reflective surfaces to a light beam.

15. The method of claim 14, wherein filling the cavity comprises pressurizing the cavity with fill gas comprising from 10 to 100% hydrogen (H) or helium (He) at a pressure of 1 atmospheres or greater.

16. The method of claim 14 wherein the package further comprises an outlet, and wherein filling the cavity comprises providing a flow of gas though the cavity from the inlet to the outlet of from 0.01 Standard Cubic Centimeters per Minute (sccm) to 10000 sccm during operation of the SLM.

17. The method of claim 14 further comprising discontinuing exposing the reflective surfaces to the light beam if pressure decreases to less than 0.1 atmospheres.

18. The SLM of claim 6 wherein each of the number of MEMS modulators further comprises a plurality of thermal sinking structures projecting from the surface of the substrate and extending through void spaces between the posts, the number of flexures and the piston of the piston layer to improve thermal management of the piston layer.

19. The system of claim 9, wherein each of the MEMS modulators comprises a two-imensional (2D) MEMS modulator including: a piston layer suspended over the surface of the substrate by posts at corners thereof, the piston layer including an electrostatically deflectable piston and a number of flexures through which the piston is coupled to the posts; a plurality of thermal sinking structures projecting from the surface of the substrate and extending through void spaces between the posts, the number of flexures and the piston of the piston layer to improve thermal management of the piston layer; a first reflective surface over a top surface of the piston; and a faceplate suspended over the piston layer, the faceplate including a second reflective surface on a top surface of the faceplate, and an aperture through which the piston exposed, wherein electrostatic deflection of the piston brings light reflected from the first reflective surface into interference with light reflected from the second reflective surface.